

ANSWER KEYS

PHYSICS

[illegible]

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	4	4	3	4	3	3	1	2	1	1	3	4	4	1	3	4	1	1	1	2
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	2	1	2	4	1	3	3	1	2	1	2	1	3	1	1	4	1	2	3	3
Q.	86	87	88	89	90															
A.	3	3	2	2	2															

BIOLOGY

[illegible]

SOLUTIONS

PHYSICS

1)

For conservative force

$$\frac{\partial F_y}{\partial x} = \frac{\partial F_x}{\partial y}$$

2) ME = 30 J

$$U_B = U_D = 35 \text{ J} > 30 \text{ J}$$

Hence particle will never reach B and D. It will oscillate between B and D.

3)

(1) → (P)

Directions of force on bucket and its displacement are same, so work done is positive.

(2) → (Q)

Directions of force of friction and displacement of body are opposite to each other, so work done is negative.

(3) → (Q)

Resistive force due to air, eventually stops the motion of pendulum.

Thus change in KE of pendulum is negative.

Since $w = \Delta KE$, the work done is negative in this case.

(4) → (P)

Directions of applied force and displacement of body are same, so work done on body due to applied force is positive.

$$4) F = ma = (2t\hat{i} + 3t^2\hat{j})\text{N}$$

$$\square m = 1 \text{ kg}$$

$$1. a = (2t\hat{i} + 3t^2\hat{j})$$

$$\frac{dv}{dt} = (2t\hat{i} + 3t^2\hat{j})$$

$$\int_0^t dv = \int_0^t (2t\hat{i} + 3t^2\hat{j}) dt$$

$$v = \left(\frac{2t^2}{2}\hat{i} + \frac{3t^3}{3}\hat{j} \right)_0^t$$

$$\vec{v} = [t^2\hat{i} + t^3\hat{j}]$$

$$P = \vec{F} \cdot \vec{v} = [2t\hat{i} + 3t^2\hat{j}] \cdot [t^2\hat{i} + t^3\hat{j}]$$

$$= [2t^3 + 3t^5] \text{ Watt}$$

5) Work done against gravity in both cases is equal = $mg\Delta h$

ratio 1 : 1

6) Using energy conservation

$$mg(R + R \sin 45^\circ) = \frac{1}{2}mv^2$$

$$v = \sqrt{2g \left(R + \frac{R}{\sqrt{2}} \right)}$$

$$v = \sqrt{2g \left(14 + \frac{14}{\sqrt{2}} \right)}$$

$$v = \sqrt{2 \times 10 \times 14 \left(1 + \frac{1}{\sqrt{2}} \right)}$$

$$v = 21.9 \text{ m/s}$$

7) Final speed of both particle is same when reached at the bottom because of energy conservation.

$$V_1 = V_2$$

$$V = u + at$$

$$t_1 = \frac{V}{g \sin 30^\circ}$$

$$t_2 = \frac{V}{g \sin 60^\circ} \quad \begin{bmatrix} a_1 = g \sin 30^\circ \\ a_2 = g \sin 60^\circ \end{bmatrix}$$

$$\frac{t_1}{t_2} = \frac{\sin 60^\circ}{\sin 30^\circ} = \frac{\sqrt{3}/2}{1/2} = \frac{\sqrt{3}}{1}$$

8)

Volume of the wind flowing through the windmill per sec = AV

Mass of the wind flowing through the windmill per sec = ρAV

Mass m of the wind flowing through the windmill

in time t = ρAVt

$$9) K.E._{\text{air}} = \frac{1}{2}mv^2 = \frac{1}{2}(\rho AVt)V^2 = \frac{1}{2}\rho AV^3t$$

10) Electric energy produced

$$= \frac{25}{100} \times \frac{1}{2}\rho AV^3t = \frac{1}{8}\rho AV^3t$$

$$11) P = \frac{E}{t} = \frac{\frac{1}{8}\rho AV^3t}{t} = \frac{1}{8}\rho AV^3$$

$$12) P \propto V^3$$

$$\frac{P'}{P} = \frac{(2V)^3}{V^3} \Rightarrow P' = 8P$$

13) Force in Negative direction

Displacement in positive direction

$$W = -ve = \Delta K.E.$$

14) Speed of particle on circular path, $u = \frac{2\pi R}{T}$
In projectile motion

$$H_{\max} = \frac{u^2 \sin^2 \theta}{2g} \Rightarrow 8R = \frac{\left(\frac{2\pi R}{T}\right)^2 \sin^2 \theta}{2g}$$

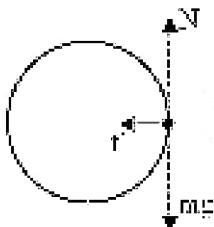
$$\frac{4gT^2}{\pi^2 R} = \sin^2 \theta \Rightarrow \sin \theta = \left(\frac{4gT^2}{\pi^2 R}\right)^{\frac{1}{2}}$$

$$\Rightarrow \theta = \sin^{-1} \left(\frac{4gT^2}{\pi^2 R}\right)^{\frac{1}{2}}$$

$$15) a_{cp} = \frac{v^2}{R} = \frac{(\sqrt{40})^2}{10} = \frac{40}{10} = 4 \text{ m/s}^2, a_T = \frac{dv}{dt} = 3 \text{ m/s}^2$$

$$a_{\text{net}} = \sqrt{a_{cp}^2 + a_T^2} = \sqrt{4^2 + 3^2} = \sqrt{25} = 5 \text{ m/s}^2$$

$$F_{\text{net}} = ma_{\text{net}} = 2 \times 5 = 10 \text{ N}$$



16)

$$N = mg \text{ \& } f = F_{cp} = \frac{mv^2}{R}$$

f is static friction as car is not moving in radial direction

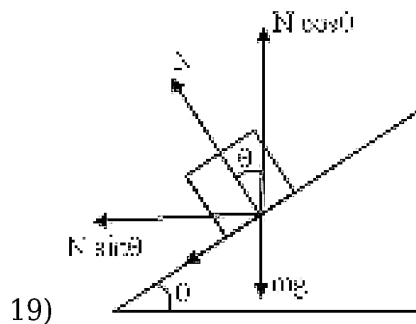
$$\text{So } f \leq f_L \Rightarrow f \leq \mu_s N$$

$$\frac{mv^2}{R} \leq \mu_s mg \Rightarrow v \leq \sqrt{\mu_s Rg}$$

17) When a stone is rotated in a horizontal circle, then centripetal force is given by $F = \frac{mv^2}{r}$

So, $F \propto v^2$ (if m & r constant) and $F \propto \frac{1}{r}$ (if m & v are constant)

18) If speed in circular motion is constant, then it is called uniform circular motion. Also in this motion only acceleration is centripetal acceleration which is directed towards centre of circle.



$$N \cos \theta = mg \dots (1) \text{ \& } N \sin \theta = \frac{mv_0^2}{r_0} \dots (2)$$

$$\frac{(2)}{(1)} \Rightarrow \tan \theta = \frac{v_0^2}{r_0 g} \Rightarrow \theta = \tan^{-1} \left(\frac{v_0^2}{r_0 g} \right)$$

20) If a particle wants to just complete vertical circle, then its speed at lowest point should be $\sqrt{5gr}$ & in that case speed at highest point should be $\sqrt{gr}$.

21) $\omega = \frac{2\pi}{T} \Rightarrow \omega \propto \frac{1}{T}$

$$\frac{\omega_{\text{minute}}}{\omega_{\text{hour}}} = \frac{T_{\text{hour}}}{T_{\text{minute}}} = \frac{12\text{hr}}{1\text{hr}} = 12 : 1$$

22) $\mu mg = m\omega_1^2 r_1 = m\omega_2^2 r_2$

$$\frac{r_2}{r_1} = \left(\frac{\omega_1}{\omega_2} \right)^2 = \left(\frac{1}{4} \right)^2 = \frac{1}{16}$$

$$\Rightarrow \frac{r_2}{9} = \frac{1}{16} \Rightarrow r_2 = \frac{9}{16} \text{ cm}$$

23) Two coils carrying current in opposite direction, hence net magnetic field at centre will be difference of the two fields.

i.e. $B_{\text{net}} = \frac{\mu_0}{4\pi} \cdot 2\pi N \left[\frac{i_1}{r_1} - \frac{i_2}{r_2} \right]$

$$= \frac{10\mu_0}{2} \left[\frac{0.2}{0.2} - \frac{0.3}{0.4} \right] = \frac{5}{4} \mu_0$$

24) $B = \mu_0 n i \Rightarrow i = \frac{B}{\mu_0 n} = \frac{20 \times 10^{-3}}{4\pi \times 10^{-7} \times 20 \times 100}$

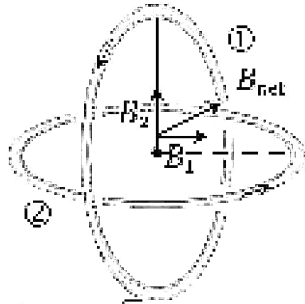
$$= 7.9 \text{ amp} \approx 8 \text{ amp}$$

25) $B = \frac{\mu_0 n i}{2r}$, As $\square = 2\pi r = n(2\pi r')$

$$r' = \frac{r}{n} \Rightarrow B' = \frac{\mu_0 n i}{2 \left(\frac{r}{n} \right)} = n^2 B$$

$$B' = n^2 B \Rightarrow B' = (4)^2 B \Rightarrow B' = 16B$$

$$26) B_1 = B_2 = B = \frac{\mu_0}{4\pi} \times \frac{2\pi i}{r}$$



$$B_{\text{net}} = \sqrt{2}B$$

$$\Rightarrow \frac{B}{B_{\text{net}}} = \frac{1}{\sqrt{2}}$$

$$27) \text{ Time period of proton } T_p = \frac{2\pi m}{qB} = 5 \mu \text{ sec}$$

By using

$$T = \frac{2\pi m}{qB} \Rightarrow \frac{T_\alpha}{T_p} = \frac{m_\alpha}{m_p} \times \frac{q_p}{q_\alpha} = \frac{4m_p}{m_p} \times \frac{q_p}{2q_p}$$

$$\Rightarrow T_\alpha = 2T_p = 10 \mu \text{ sec.}$$

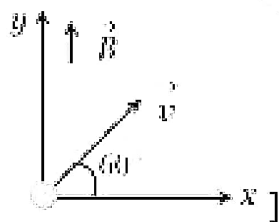
$$28) r = \frac{\sqrt{2mK}}{qB} \Rightarrow r_p = r_d$$

$$\Rightarrow \frac{\sqrt{2m_p k_p}}{q_p B} = \frac{\sqrt{2m_d k_d}}{q_d B}$$

$$\Rightarrow m_p k_p = m_d k_d \Rightarrow k_p = 2k_d$$

$$\Rightarrow K_p = 2 \times 50 = 100 \text{ keV}$$

$$29) \text{ Path of the proton will be a helix of radius } r = \frac{mv \sin \theta}{qB}$$



(where θ = Angle between $\vec{B}$ and $\vec{v}$)

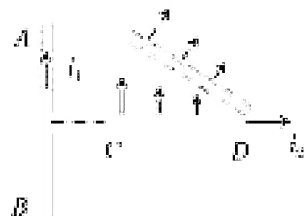
$$\Rightarrow r = \frac{1.67 \times 10^{-27} \times 2 \times 10^6 \times \sin 30^\circ}{1.6 \times 10^{-19} \times 0.104} = 0.1 \text{ m}$$

$$\text{Time period } T = \frac{2\pi m}{qB} = \frac{2\pi \times 1.67 \times 10^{-27}}{1.6 \times 10^{-19} \times 0.104}$$

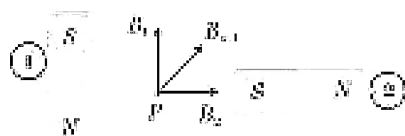
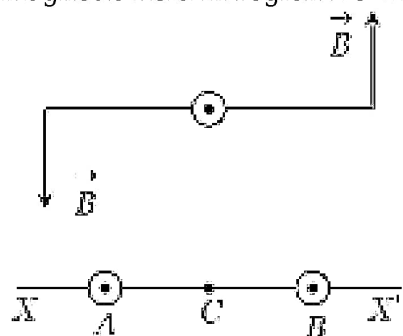
$$= 2\pi \times 10^{-7} \text{ sec}$$

$$30) \text{ Using } eE = evB \Rightarrow E = vB = 5 \times 10^6 \times 0.02 = 10^5 \text{ Vm}^{-1}$$

31) Since the force on the rod CD is non-uniform it will experience force and torque. From the left hand side it can be seen that the force will be upward and torque is clockwise.



32) If the current flows out of the paper, the magnetic field at points to the right of the wire will be upwards and to the left will be downward. Now magnetic field at C, is zero. The field in the region BX' will be upwards (+ve) because all points lying in this region are to the right of both the wires. Similarly, magnetic field in the region AX will be downwards (-ve). The field in the region AC will be upwards (+ve) because points are closer to A compared to B. Similarly magnetic field in region BC will be downward (-ve). Graph (2) satisfies all these conditions.



33)

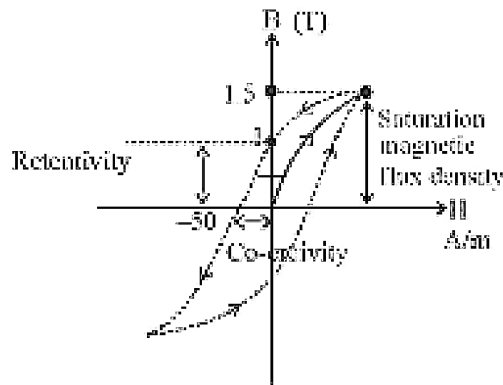
$$34) \quad T = 2\pi \sqrt{\frac{l}{MB}} \Rightarrow \frac{T_1}{T_2} = \sqrt{\frac{M_2}{M_1}}$$

$$\Rightarrow \frac{M_1}{M_2} = \frac{T_2^2}{T_1^2} = \frac{(60/15)^2}{(60/10)^2} = \frac{4}{9}$$

35) Every point on line AB will be equidistant from X and Y-axis. So magnetic field at every point on line AB due to wire 1 along X-axis is equal in magnitude but opposite in direction to the magnetic field due to wire along Y-axis. Hence B_{net} on AB = 0

$$36) \quad F = mB \Rightarrow F = \frac{M}{L} \times B$$

$$\Rightarrow 6 \times 10^{-4} = \frac{3}{L} \times 2 \times 10^{-5} \Rightarrow L = 0.1 \text{ m}$$



37)

Retentivity = 1.0 T

Co-ercivity = 50 A/m

Saturation magnetic flux density = 1.5 T

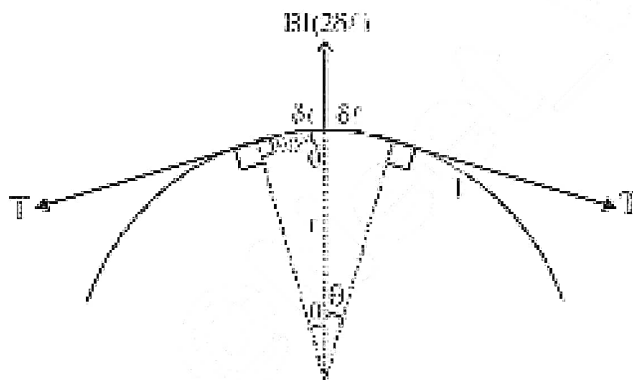
38)

$$W_{\text{ext.}} = MB(\cos\theta_1 - \cos\theta_2)$$

$$W_{\text{ext.}} = 20 \times 5 \times 10^{-4} (\cos 60^\circ - \cos 0^\circ)$$

$$W_{\text{ext.}} = -5 \times 10^{-3}$$

$$W_{\text{mag.}} = -W_{\text{ext.}} = 5 \times 10^{-3} \text{ J}$$



39)

$$2T \sin\theta = BI(2\delta l)$$

$$\Rightarrow T \sin\theta = BI\delta l$$

$$\Rightarrow T \times \frac{\delta l}{r} = BI\delta l$$

$$\Rightarrow T = BIr = \frac{BIL}{2\pi}$$

40) Conductor A attracts conductor C towards A

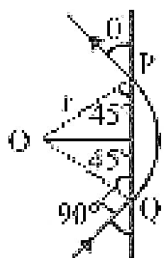
□ Force towards A

$$\begin{aligned} & \frac{\mu_0 I_1 I_2}{2\pi r} = \frac{\mu_0 2 \times 3 \times 1}{2\pi \times 0.05} = \frac{3\mu_0}{\pi} \times 20 \\ & = \frac{60 \times 4\pi \times 10^{-7}}{\pi} = 240 \times 10^{-7} \text{ N} \end{aligned}$$

$$\begin{aligned} \text{Again force towards B} &= \frac{\mu_0}{2\pi} \times \frac{3 \times 4 \times 1}{0.08} = \frac{3\mu_0}{\pi} \times 25 \\ &= \frac{3 \times 4\pi \times 10^{-7}}{\pi} \times 25 = 300 \times 10^{-7} \text{ N} \end{aligned}$$

- Resultant force towards
 $B = (300 - 240)10^{-7} \text{ N}$
 $= 0.6 \times 10^{-5} \text{ N}$

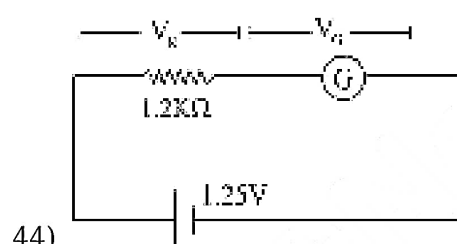
41) $\frac{PQ}{2} = r \cos 45^\circ$



$PQ = 2r \cos 45^\circ \quad r = \frac{mv}{qB}$

42) Using formula for Half deflection method $G = \frac{RS}{R-S}$
 $G = \frac{1000 \times 91}{909} \cong 100\Omega$

- 43) The figure of merit of galvanometer is defined as the current required to produce a unit deflection on its scale.



44) Current through galvanometer without shunt $I = 25 \mu\text{A/div} \times 40 \text{ div} = 1000 \mu\text{A} = 1\text{mA}$
 $V_R = IR \Rightarrow V_R = 10^{-3} \times 1200 = 1.2 \text{ V}$
Voltage across galvanometer $V_G = V - V_R = 1.25 - 1.2 = 0.05 \text{ V}$

Resistance of Galvanometer $G = \frac{V_G}{I_g} = \frac{0.05}{0.001} = 50\Omega$

As $G = \frac{RS}{R-S}$, so
 $S = \frac{GR}{G+R} = \frac{(50)(1200)}{1250} = 48\Omega$

45) Figure of merit (K) = $\frac{\text{Current}}{\text{deflection}} = \frac{i}{\theta} = \frac{3 \times 10^{-3} \text{ A}}{24 \text{ degree}}$
 $= \frac{3 \times 10^{-3} \text{ A}}{12 \text{ div}} = \frac{1}{4} \times 10^{-3} \frac{\text{A}}{\text{div}}$
 $= 2.5 \times 10^{-4} \text{ A/div}$

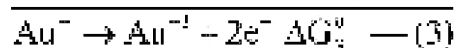
$$46) \pi \propto M \quad \pi_{\text{final}} = \frac{\pi_1 V_1 + \pi_2 V_2}{V_1 + V_2} = \frac{1.64V + 2.46V}{2V}$$

$$\pi_{\text{final}} = 2.05 \text{ atm}$$

47) Van't Hoff factor 'i' decreases ; freezing point increases.

48) Van't Hoff factor 'i' for urea = 1 and for NaCl = 2

$$\pi = iCRT$$



$$49) \text{Eq. (3)} = \text{Eq. (1)} - \text{Eq. (2)}$$

$$50) P_T = 180X_A + 90 \quad \text{--- (1)}$$

$$P_T = (P_A^0 - P_B^0) X_A + P_B^0 \quad \text{--- (2)}$$

From equation (1) & (2)

$$P_B^0 = 90 \text{ mm Hg}$$

$$P_A^0 - P_B^0 = 180$$

$$\square P_A^0 = 180 + P_B^0 = 180$$

$$P_A^0 = 270 \text{ mm Hg}$$

$$\Rightarrow \frac{P_A^0}{P_B^0} = \frac{270}{90} = 3$$

51)

$$\frac{(\Delta T_f)_1}{(\Delta T_f)_2} = \frac{i_1 K_f m_1}{i_2 K_f m_2}$$

$$\frac{2.15}{(\Delta T_f)_2} = \frac{i_1 K_f \cdot \frac{5}{342} \times \frac{1000}{95}}{i_2 K_f \cdot \frac{5}{180} \times \frac{1000}{95}}$$

$$\frac{2.15}{(\Delta T_f)_2} = \frac{180}{342} \Rightarrow (\Delta T_f)_2 = 4.085$$

$$\Delta T_f = T_f^0 - T_f$$

$$T_f = 273.15 - 4.085$$

$$T_f = 269.065 \text{ K}$$

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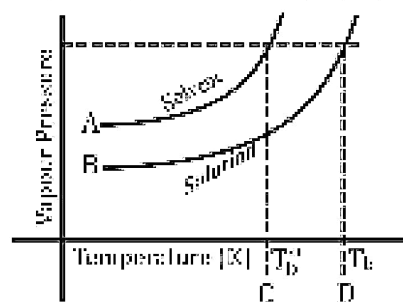
Hence, option (3) is correct.

52) By definition isotonic solutions have same osmotic pressure at same temperature hence when these are separated by spm, then there is no osmosis.

53)

Solution/Explanation/Calculation :-

In given graph identify A, B, C and D.



Hence, option (2) is correct.

(A-solvent) (B-solution) (C- T_b^0), (D- T_b)

54)

$$P = K_H X = K_H \times \frac{n_{\text{solute}}}{n_{\text{solute}} + n_{\text{solvent}}}$$

$$P = K_H X_{\text{gas}} \quad \frac{3.2 \times 10^5}{}$$

$$\Rightarrow X_{\text{gas}} = 1.6 \times 10^{-3}$$

$$X_{\text{gas}} = 2 \times 10^{-3}$$

as mole fraction is very small

$$\text{hence } X_{\text{gas}} \approx \frac{n_{\text{gas}}}{n_{\text{water}}}$$

$$2 \times 10^{-3} = \frac{n_{\text{CO}_2}}{(500/18)}$$

$$n_{\text{CO}_2} = 2 \times 10^{-3} \times \frac{500}{18}$$

$$n_{\text{CO}_2} = \frac{1}{18}$$

$$\Rightarrow W_{\text{CO}_2} = 2.44\text{g}$$

55)

As concentration increases V.P. decreases.

In given graph solution-I is less concentrated and solution-III is more concentrated because V.P. of solution-I is less decreases as compared to solution-III. So order to concentrated solution is $I < II < III$.

Final Answer : (1) $I < II < III$

56) **Explain Question :**

Which is isotonic with 15% $\frac{w}{w}$ glucose solution.

Concept :

Isotonic solution

Solution :

for glucose $i = 1$

$$\pi = iCRT$$

$$\pi_{\text{glucose}} = \frac{1 \times \frac{\%w/w \times 10 \times d}{M_w}}{RT}$$

$$= 1 \times \frac{15 \times 10 \times 1.2}{180} \times RT$$

$$\pi_{\text{glucose}} = RT$$

isotonic solution i.e same osmotic pressure

$$0.5M \Rightarrow \pi_{\text{NaCl}} = iCRT = 2 \times 0.5 RT = RT$$

$$1M \Rightarrow \pi_{\text{CaCl}_2} = iCRT = 3 \times 1RT = 3RT$$

$$1M \Rightarrow \pi_{\text{CaCl}_2} = iCRT = 3 \times 1 \times RT = 3RT$$

$$0.5 \Rightarrow \pi_{\text{urea}} = iCRT = 1 \times 0.5 RT = 0.5RT$$

Final Answer option (3)

OR

$$[\text{Glucose}] = \frac{15 \times 1.2 \times 10}{180} = 1 \text{ M}$$

57)

For $K_4[\text{Fe}(\text{CN})_6]$

$$i = 1 + (5 - 1)\alpha$$

$$i = 1 + 4x \text{ (D.O.D = x)}$$

$$\frac{m_{\text{Normal}}}{m_{\text{Abnormal}}} = \frac{1}{i}$$

$$m_{\text{Abnormal}} = \frac{m_{\text{Normal}}}{i} = \frac{m}{(1 + 4x)}$$

58)

Explanation

A. **$\text{C}_6\text{H}_5\text{Cl}$ - $\text{C}_6\text{H}_5\text{Br}$ (Chlorobenzene - Bromobenzene):**

A. Both are aryl halides with similar structures and van der Waals forces. Thus they will form an ideal solution.

B. **C_6H_6 - $\text{C}_6\text{H}_5\text{CH}_3$ (Benzene - Toluene):**

A. Both are aromatic hydrocarbons with very similar structures and van der Waals forces. They will form an ideal solution.

C. **Hexane-heptane:**

A. Both are nonpolar alkanes with very similar structures and van der Waals forces. They will form an ideal solution.

Since all the options have similar intermolecular forces, they will form ideal solutions.

Answer option 4, (All of the above).

$$59) \Delta T_b = i K_b \times m$$

$$= 3 \times 0.52 \times \frac{13.44}{134.4} \times \frac{1}{1} = 0.156$$

60) B.P. $\propto$ moles of non-volatile solute $\propto i c$

61) Only solvent molecules can pass through SPM so only dilution is possible.

$$62) P_g = K_H \cdot \frac{n_g}{n_{H_2O}}$$

$$1 = K_H \cdot \frac{0.02}{28} \times \frac{18}{1000}$$

$$K_H = \frac{0.02 \times 18}{28 \times 1000} = 7.7 \times 10^4 \text{ atm}$$

63)

$$\text{Solubility} \propto \frac{1}{K_H}$$

$$64) \frac{P^0 - P_s}{P_s} = \frac{n_{\text{solute}}}{n_{\text{solvent}}}$$

$$\frac{P^0 - 20}{20} = \frac{6}{M} \times \frac{18}{180} \dots\dots\dots(1)$$

$$\frac{P^0 - 20.02}{20.02} = \frac{6}{M} \times \frac{18}{198} \dots\dots\dots(2)$$

$$P^0 = 20.22 \text{ torr.}$$

65) Fe^{+3} ions are absorb on surface of $\text{Fe}(\text{OH})_3$.

66) $\text{Fe}(\text{OH})_3$ is lyophobic sol can be prepared by special method.

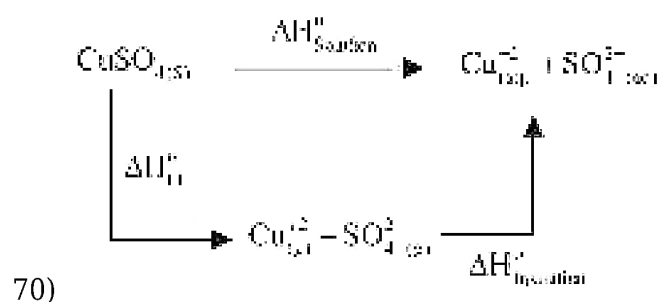
67)

$$Y_B P_s = X_B P_B^0$$

$$68) \Lambda_m = \Lambda_m^0 - AC^{\frac{1}{2}}$$

69)

In Lyophilic sol. attraction present between DP and DM.



71) pH range for titration is 6 to 11.

72)

In option 3 both are strong acid & base therefore has maximum heat of neutralisation.

73) SRP ↓ Reducing power ↑

74) (P) Water oxidise at anode and Cu^{+2} reduce at cathode

(Q) Cu oxidise at anode and Cu^{+2} reduce at cathode

(R) Cl^- oxidise at anode and H_2O reduce at cathode

(S) Na^+ reduce at cathode and produce amalgam.

75)

Given Data:

$$E^\circ_{\text{cell}} < 0$$

A. $\Delta G^\circ = -nFE^\circ_{\text{cell}}$

B. $\Delta G^\circ = -RT \ln K_{\text{eq}}$

76)

Explanation:

This is a concentration cell involving chlorine gas (Cl_2) at different partial pressures. The reaction occurs as: $\text{Cl}_2(P_2) + 2\text{Cl}^- \rightleftharpoons \text{Cl}_2(P_1) + 2\text{Cl}^-$

Concept:

Since it is a gas concentration cell, the spontaneity of the reaction is determined by the Nernst equation: $E = - (0.0591/n) \times \log(P_1/P_2)$

For a spontaneous reaction, $E > 0$, which happens when $P_2 > P_1$.

Answer Option 2, ($P_2 > P_1$)

$$77) \kappa = \frac{1}{\rho} = \frac{1}{R} \left(\frac{\ell}{A} \right) = \frac{1}{50} \times \frac{2.1}{4.2} = \frac{1}{100} \text{ and}$$
$$\Lambda_{\text{eq.}} = \frac{\kappa \times 1000}{N} = \frac{1}{100} \times \frac{1000}{1}$$

$$= 10 \text{ ohm}^{-1} \text{ cm}^2 \text{ eq}^{-1}$$

78) **Question Explanation :-** To determine degree of dissociation of NH_4OH (α)

Given Data:-

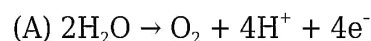
$$\Lambda_m = 9.54 \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$$

$$\Lambda_m^\infty = 238 \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$$

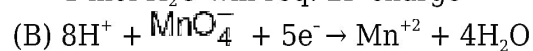
$$\text{Solution explanation} - :- \quad \alpha = \frac{\Lambda_m}{\Lambda_m^\infty} \Rightarrow \alpha = \frac{9.54}{238} \times 100$$

final Answer :- Option (3) 4.008 %

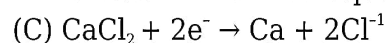
79)



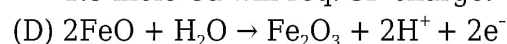
1 mol H_2O will req. 2F charge



1 mole MnO_4^- will req. 5F charge.



1.5 mole Ca will req. 3F charge.



1 mole FeO will req. 1 F charge.

(A)-(II), (B)-(IV), (C)-(I), (D)-(III)

80) $\lambda_{\text{eq}}(\text{A}_3\text{B}_2) = 200 + 400$
 $= 600$

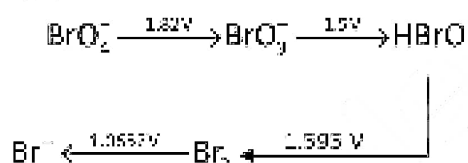
81)

Concept:

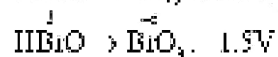
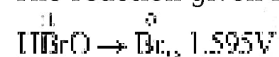
Disproportionation Reaction and Reduction potential.

Explanation:-

Given



The reaction given in the question is as follows.



E°_{cell} for the disproportionation of HBrO,

$$E^\circ_{\text{cell}} = E^\circ_{\text{reduction}} + E^\circ_{\text{oxidation}}$$

$$= 1.595 - 1.5$$

$$= 0.095\text{V} = +\text{ve}$$

Final Answer Option for the given question is (3) HBrO

82) NCERT Pg#86 to 89

83) $\Delta G^\circ = -nFE^\circ$

$$\Delta G^\circ = -2 \times 96500 \times 1.211\text{J}$$

$$\Delta G^\circ = -233.7\text{kJ}$$

84) $\Lambda_m = \frac{\kappa \times 1000}{M}$

$$\& \kappa = G.G^*$$

85)

Faraday Ist law,

$$w_{Al} = \frac{E_{Al}}{9} \times i \times t$$

$$W_{Al} = \frac{96500}{\text{At. wt } 27} \times 4 \times 10^4 (6 \times 60 \times 60)$$

$$E_{Al} = \frac{v.f.}{3} = 9$$

$$W_{Al} = 8.1 \times 10^4 g$$

86)

$$E_{\text{cell}} = (E_{RP})_{\text{cathode}} - (E_{RP})_{\text{Anode}}$$

$$= (E_{RP})_{\text{Cathode}} + (E_{OP})_{\text{Anode}}$$

$$(\square E_{RP} = - E_{OP})$$

$$= (E_{OP})_{\text{Anode}} - (E_{OP})_{\text{Cathode}}$$

$$87) \alpha = \frac{\Lambda_m^0}{\Lambda_m^\infty} = \frac{48}{344 + 40} = 0.125$$

$$\text{For } CH_3COOH \quad K\alpha = \frac{C \alpha^2}{1 - \alpha}$$

$$\square K_a = \frac{0.008 \times (0.125)^2}{1 - 0.125} = 1.4 \times 10^{-4}$$

88)

In galvanic cell, electron more from anode to cathode in external circuit so current flows from cathode to anode. From representation Cu to Ag.

$$89) \Delta G_3^0 = \Delta G_1^0 + \Delta G_2^0$$

$$n_3 \times E_3^0 = n_1 E_1^0 + n_2 E_2^0$$

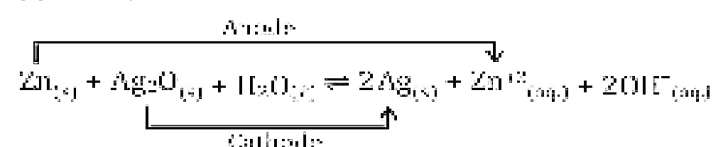
$$2 \times E_{Cu^{2+}/Cu}^0 = 1 \times E_{Cu^{2+}/Cu^+}^0 + 1 \times E_{Cu^+/Cu}^0$$

$$E_{Cu^{2+}/Cu}^0 = \frac{1 \times 0.15 + 1 \times 0.50}{2}$$

$$E_{Cu^{2+}/Cu}^0 = 0.325 V$$

90)

Cell rxn :



$$E_{\text{Cell}}^{\circ} = [\text{SRP}]_{\text{C}} - [\text{SRP}]_{\text{A}}$$

$$= 0.34 - (-0.76) = 1.1\text{V}$$

BIOLOGY

91) NCERT-XII, Pg # 83(E), 91(H)

92)

NCERT XII Pg. # 60, 61

93)

NCERT-XII, Pg. # 57

94)

It is the diagrammatic representation of a test cross to find out the genotype of dominant organisms.

95) **The correct answer is: 1. 6 and 4**

The ABO blood group system is determined by the presence of A, B, and O alleles. The possible genotypes are:

- A. AA
- B. AO
- C. BB
- D. BO
- E. AB
- F. OO

The possible phenotypes based on these genotypes are:

- A. A (from AA or AO)
- B. B (from BB or BO)
- C. AB (from AB)
- D. O (from OO)

So, there are 6 possible genotypes and 4 possible phenotypes.

96)

NCERT Pg. No.# 76

97)

NCERT XII, Pg # 63

98) NCERT-XII Pg. # 62

99) NCERT-XII, Pg # 74

100) NCERT-XII, Pg. # 72, 73

101)

NCERT XII, Pg # 67

102)

NCERT_XII_Pg. No. # 70

103)

Module

104)

NCERT XII, Pg # 60

105)

NCERT XII, Pg # 67

106)

NCERT XII, Pg # 67

107)

NCERT XII, Pg # 69

108)

NCERT_XII_Pg. No. # 64

109)

NCERT_XII_Pg. No.# 75

110)

NCERT XII, Pg # 68

111)

NCERT_XII_Pg. No.# 55

112)

NCERT XII, Pg # 63

113)

NCERT XII, Pg # 73

114)

NCERT-XII_Pg. No.# 73

115)

NCERT_XII_Pg. No.#76

116)

NCERT XII, Pg # 66

117)

Module

118)

NCERT-XII_Pg. No. # 62

119)

NCERT_Pg. No. # 74, 76

120)

NCERT XII, Pg # 69

121)

NCERT-XII, Pg # 83

122) NCERT_Pg # 69

123) XIIth New NCERT Pg. # 68

124)

NCERT XII, Pg # 72

125)

NCERT XII, Pg # 72

126)

NCERT-XII_Pg. No.# 71

127)

NCERT XII, Pg # 68

128)

NCERT XII, Pg # 57, 58

129) Statements (ii) and (iii) are correct.

130)

NCERT pg no.73

131) NCERT XII Pg. # 83

132) NCERT XII Pg. # 54

133) NCERT-XII Pg. # 64

134)

NCERT XII, Pg # 60

135)

NCERT XII, Pg. # 60,62

136) NCERT XII, Page # 183

137) NCERT XII Pg. 184

138)

NCERT XII, Page # 182

139)

NCERT XII, Page # 161, 165, 168, 169

140)

NCERT XII, Page # 167

141) NCERT XII Pg # 202

142)

NCERT XII, Page # 172

143)

NCERT XII, Page # 165, 171-172

144)

NCERT XII, Page # 179

145) NCERT XII, Pg # 182

146)

NCERT Pg. # (E)-185 (H)-203

147)

NCERT XII, Page # 182

148)

NCERT XII, Page # 179

149) NCERT Pg.# 209

150)

NCERT XII, Page # 183

151) NCERT-XII, Pg # 194 (E)

152) NCERT XII Pg # 170

153)

NCERT XII Pg # 200

154) **Explanation:** The Roman numeral in the name of restriction enzymes (like EcoRI) indicates the order in which the enzyme was isolated from its bacterial strain. The "Eco" part refers to the genus of the bacterium *Escherichia coli*, and the "R" refers to the specific strain of the bacterium. The Roman numeral denotes the sequence of isolation.

The correct answer is 2.

Order in which the enzymes were isolated from the strain of bacteria

155)

NCERT XII, Page # 171-172

156) NCERT-XII, Page # 209 (12.1)

157)

NCERT pg.no.195

158)

NCERT-XII, Pg # 172, 173

159) NCERT-XII, Pg. # 196, 197

160) NCERT, Pg. # 168 & 172

161)

NCERT XII, Page # 174

162) NCERT Pg. # 197, 198

163)

NCERT XII, Page # 170

164) NCERT Pg.#180,182,184,183

165)

NCERT XII, Page # 180

166)

NCERT XII, Page # 172

167)

NCERT XII, Page # 172

168)

NCERT XII, Page # 169

169) NCERT Pg.#170,173,174,185,168

170)

NCERT XII, Page # 169

171)

NCERT XII, Page # 171

172)

NCERT, Class-XII Page No. 182 (E), 199 (H)

173)

NCERT XII, Page # 166

174)

The correct answer is option 1: Hind III, Hind III.

175)

NCERT XII, Page # 164-166, 170

176)

NCERT Pg.# 165, 166

177) Question Explanation :

The question is asking about two statement

Concept :

This question is based on vector

Solution :

Statement I is true: Normal E.coli cells lack antibiotic resistance genes like ampicillin, chloramphenicol, tetracycline, kanamycin.

Statement II is true: Blue colonies appear when the plasmid does not have an insert (due to functional β -galactosidase acting on chromogenic substrate). Correct Answer: (1) Both Statement I and II are correct.

Final Answer :

Option (1)

178)

NCERT-XII, Pg. # 214

179)

NCERT-XII Pg. # 210

180) NCERT-XII Pg. # 232